

CHAPLEAU, ON, Canada

WMO: 716420

Lat: 47.820N

Lon: 83.347W

Elev: 447

StdP: 96.07

Time Zone: -5.00 (NAE)

Period: 96-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.5	-28.9	-35.8	0.1	-31.2	-32.1	0.2	-27.3	8.5	-6.2	7.6	-7.4	0.9	320	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	28.2	19.8	26.8	19.0	25.0	17.7	21.8	26.7	20.8	25.7	19.7	24.2	4.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	15.6	24.3	19.0	14.6	23.3	18.0	13.7	22.2	65.9	27.0	62.1	25.7	58.3	24.3	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.2	6.4	DB	-37.7	31.0	3.2	1.9	-40.0	32.3	-41.9	33.5	-43.7	34.5	-46.0	35.9	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	-14.8	-13.1	-6.7	1.9	10.0	15.7	17.6	16.4	12.2	5.0	-2.4	-10.8	(6)	
	DBStd	12.81	7.80	6.92	7.51	5.46	5.14	4.13	3.32	3.44	4.63	5.02	6.00	6.98	(7)	
	HDD10.0	3472	769	647	518	248	64	6	0	1	29	172	374	646	(8)	
	HDD18.3	5801	1027	880	775	493	262	99	54	80	190	413	623	904	(9)	
	CDD10.0	796	0	0	1	4	66	176	236	199	96	18	1	0	(10)	
	CDD18.3	82	0	0	0	0	4	18	32	21	6	1	0	0	(11)	
	CDH23.3	787	0	0	0	3	72	205	283	173	48	2	0	0	(12)	
	CDH26.7	158	0	0	0	0	18	43	57	33	8	0	0	0	(13)	
(14) Wind	WSAvg	3.0	2.7	3.0	3.1	3.3	3.4	3.1	2.9	2.8	2.9	2.9	2.9	2.7	(14)	
(15) Precipitation	PrecAvg	874	52	41	48	65	79	93	87	89	107	86	78	53	(15)	
	PrecMax	1174	97	95	79	128	149	160	165	155	191	159	144	115	(16)	
	PrecMin	633	15	13	2	22	37	37	32	47	54	33	38	14	(17)	
	PrecStd	138	21	21	18	29	29	31	35	30	32	34	31	22	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	5.6	16.2	22.1	29.0	30.0	30.2	29.8	27.9	22.8	13.9	4.9	(19)	
		MCWB	2.1	3.3	9.4	12.0	19.6	20.3	21.3	21.1	19.6	16.2	10.1	4.6	(20)	
	2%	DB	1.1	2.2	10.8	17.8	25.0	27.9	28.1	27.1	24.1	18.9	10.1	2.2	(21)	
		MCWB	0.6	0.8	5.8	9.5	15.6	18.9	20.2	19.7	18.0	13.7	7.9	1.4	(22)	
	5%	DB	-0.9	0.0	7.0	14.2	22.1	25.8	26.2	25.1	22.0	15.9	7.5	0.9	(23)	
		MCWB	-1.4	-1.1	3.2	7.2	13.2	17.4	18.9	18.5	16.8	12.3	5.6	0.3	(24)	
	10%	DB	-3.0	-2.2	4.2	11.1	19.4	23.8	24.8	23.1	19.9	13.0	5.2	-0.3	(25)	
		MCWB	-3.6	-3.2	1.2	5.4	11.9	16.3	18.0	17.4	15.9	10.5	4.0	-0.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	3.8	14.5	14.5	21.2	22.0	23.4	22.9	21.3	17.2	11.3	4.7	(27)	
		MCDB	2.8	5.5	18.4	21.1	27.3	27.8	27.7	27.2	25.5	21.3	14.0	4.8	(28)	
	2%	WB	0.8	1.3	7.4	11.3	18.0	20.9	22.1	21.4	19.6	15.4	8.9	1.8	(29)	
		MCDB	1.1	2.5	10.5	16.7	24.3	26.2	27.1	25.6	23.7	18.7	10.7	2.2	(30)	
	5%	WB	-1.3	-0.5	4.9	9.0	15.6	19.7	20.9	20.1	18.4	13.3	6.1	0.4	(31)	
		MCDB	-0.6	0.6	7.7	14.6	20.8	24.6	26.1	24.4	21.9	16.4	7.5	0.7	(32)	
	10%	WB	-3.4	-2.9	2.3	7.0	13.7	18.4	19.7	19.0	16.9	11.1	4.2	-0.8	(33)	
		MCDB	-2.9	-2.0	5.1	11.8	19.1	22.9	24.0	23.1	20.5	13.9	5.6	-0.3	(34)	
(35) Mean Daily Temperature Range	MDBR		11.0	12.2	12.9	11.9	12.7	11.8	11.1	11.1	10.5	8.4	7.0	9.0	(35)	
	5% DB	MCDBR	8.3	11.1	14.2	17.2	16.6	14.9	13.6	13.3	13.2	13.0	9.7	7.9	(36)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.260	0.268	0.295	0.334	0.352	0.367	0.382	0.362	0.356	0.319	0.284	0.267	(40)	
	taud		2.296	2.273	2.311	2.370	2.426	2.396	2.365	2.422	2.439	2.527	2.511	2.317	(41)	
	Ebn at Noon		824	900	927	919	911	894	875	880	852	836	803	764	(42)	
	Edh at Noon		73	93	107	112	110	115	117	106	95	74	60	63	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.26	2.32	3.78	4.75	5.34	5.69	5.58	4.66	3.49	2.07	1.10	0.87	(44)	
	RadStd		0.22	0.32	0.30	0.42	0.41	0.39	0.42	0.33	0.27	0.27	0.11	0.10	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page